

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

6 September 2026

Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Its demonstrated results establish what coordination leaves undetermined: increasing coupling, decreasing diffusion and a matched common drive can produce identical stationary phase distributions; equal phases permit incompatible local states; and a plasticity control preserves coherence while losing alignment with a specified environmental template. The conditional construction glues compatible spatial measures into one state and makes that state the unique fixed point of a contracting reconstruction map. Lean 4 checks the composition under eight connecting hypotheses and additional structures, with compatibility and accuracy at the glued state supplied as premises. We propose the glued state as a formal correlate of conscious unity and its accurate internal reconstruction as a correlate of a minimal self. The physical hypothesis is that changing extracellular geometry increases field-mediated coupling during awakening. A noisy mean-field phase model gives a square-root stationary onset, but finite ramps delay the response and the distributional

equivalence requires measurements that distinguish recovery mechanisms. Testing the account further requires identifying neural content and self-encoding independently, then determining whether coordination maintains their agreement and whether these relations matter for unified experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Yet even successful coordination leaves a question: what makes the coordinated activity a description of one experienced situation, with a perspective on that situation?

We propose an account with three requirements. *Coherence* concerns whether coupling sustains collective activity. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting global state supports an accurate model of itself. The latter two express commitments about consciousness whose motivation and possible dissociations we develop below.

Consider the experience of reaching for a cup. A description of the cup's location and a description of the hand's relation to it can share information about where the cup is while contributing different contents to the episode.

19 Synchronous activity in the two processes would leave any disagreement
20 about that shared quantity unresolved. The formal counterexample isolates
21 this problem using spatial mass profiles at a common phase that disagree
22 on an overlap. We return to the cup example to ask what agreement would
23 mean for perceptual content, and what an internal representation would have
24 to add to place the perceiver in that situation. It is an illustration of the
25 modelling task, not a neural model implemented here.

26 Awakening gives the physical requirement a concrete setting. We hypoth-
27 esize that changing extracellular geometry slowly increases effective coupling
28 through a synchronization threshold. Under stationary branch tracking, a
29 noisy phase model fixes a square-root onset of coherence. The experiment
30 faces two distinct obstacles. Finite ramp speeds delay the response, so a
31 fitted onset exponent depends on the drive. Moreover, successive stationary
32 phase distributions can be identical under increasing coupling, decreasing
33 diffusion or a suitably matched common drive. Recovery must therefore be
34 measured in a way that identifies dynamics and distinguishes causes, rather
35 than inferred from the shape of a coherence trace alone.

36 A plasticity control tests another route to representation. Its squared-
37 drift objective rewards equalizing drift, which in the chosen frequency-cluster
38 model favors connections across the clusters that define the diagnostic tem-
39 plate. Coherence survives while alignment with that template declines. To-
40 gether with the overlap counterexample, this result establishes a limit on
41 what the coordination model explains. Compatible local content needs its

own mechanism and observable.

The contribution has three levels. The separation results establish limits on what coordination explains. The conditional construction gives a unique global state from specified compatible spatial measures and makes it the unique fixed point of a contracting map assumed to reconstruct that state. Lean 4 checks the composition, and toy witnesses establish joint satisfiability of its assumptions. The biological interpretation proposes that the glued state and its internal reconstruction capture conscious unity and a minimal self. This last step requires identifying meaningful neural descriptions and explaining how coordination maintains their agreement. Figure 1 locates the proposed connections, and Section 6 states the physical and thermodynamic assumptions.

2 Coherence: the coupling model

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are

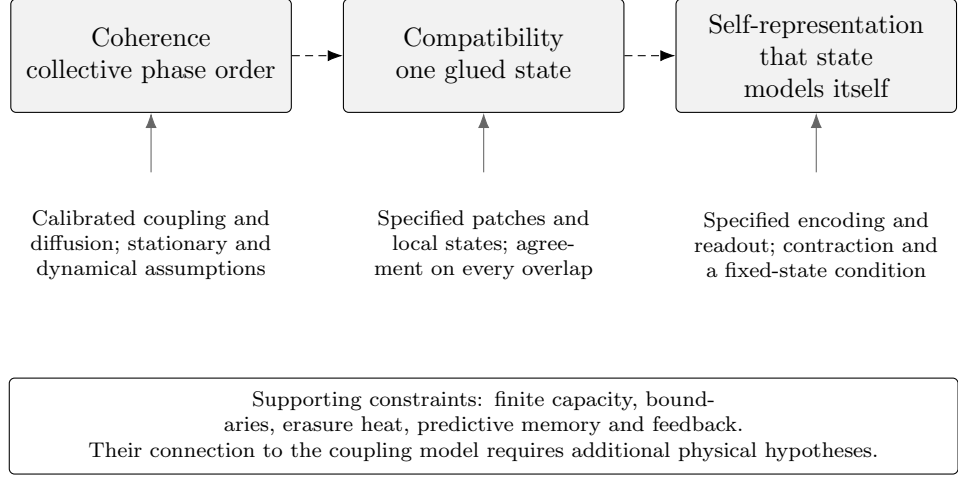


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

60 separate parameters.

61 For identical frequencies in their rotating frame, the stationary ansatz is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

62 where I_n are modified Bessel functions and ψ is the collective phase. Lean
 63 verifies normalization, the circular order parameter and self-consistency for
 64 this density. At $K \leq K_c = 2D$, the only nonnegative self-consistent order
 65 parameter is zero. At $K > K_c$, exactly one positive solution coexists with

66 zero; it emerges continuously and increases strictly with K . Near threshold,

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

67 These results characterize the stationary branch (Fig. 2). Connecting it
 68 to finite trajectories requires a dynamical mean-field limit and dynamical
 69 selection, neither proved here. Propagation of chaos concerns the first of
 70 these steps (Sznitman, 1991); exchange symmetry alone does not establish
 71 independence, especially when realizations have an unpinned collective phase.

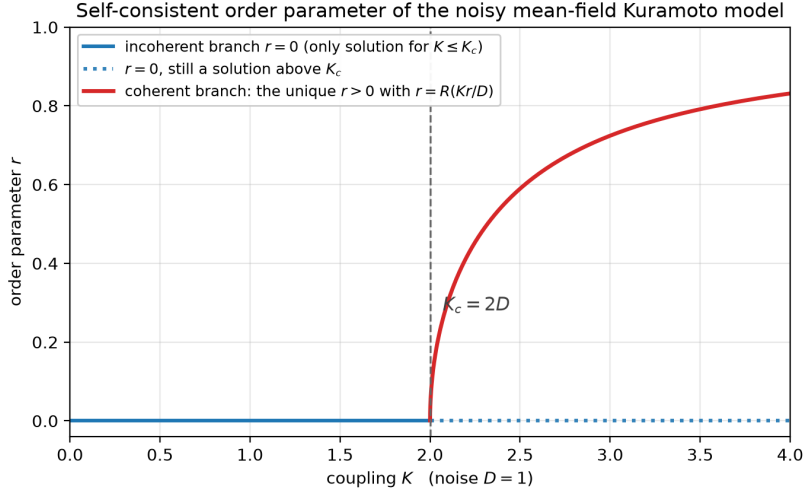


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

72 A spatial version replaces the finite sum by an integral against a kernel
 73 $K(x, y)$. The substrate measure has total mass one, making the mean-field

74 strength a kernel average. Applying the scalar threshold to a heterogeneous
 75 or signed kernel requires a justified reduction to that mean-field description.

76 **2.2 Physical identification and calibration**

77 The cortical hypothesis identifies part of the effective coupling with endoge-
 78 nous extracellular electromagnetic interactions. Fields can modulate spike
 79 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field
 80 models have been proposed for coordinating distributed neural information
 81 (Pinotsis & Miller, 2022; McFadden, 2020). The quantitative question is
 82 whether that interaction supplies sufficient coupling relative to diffusion.
 83 Changes in phase organization under propofol (Bastos et al., 2021) moti-
 84 vate studying this regime, but do not measure the coupling itself.

85 The physical field predicate requires a jointly continuous kernel, a nor-
 86 malized substrate, a mean-field strength equal to the spatial average, and no
 87 single site providing more than half that strength. Regular mesh refinement
 88 establishes convergence of a scalar coupling-energy quadrature; constructing
 89 the kernel and relating it to trajectories are separate modelling tasks. The
 90 formal witnesses supply toy instances of the predicate.

91 Calibration is essential because a spike-time shift is not a coupling rate.
 92 An estimate proportional to $N\Delta tf$, using population size, timing shift and
 93 oscillation frequency, is dimensionless. A conversion of the form

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

94 requires a rate γ , where s is the timing shift per unit field and E the field
 95 amplitude. This rate specifies how induced timing shifts persist and accumu-
 96 late into phase drift. Supplemental Material explains why the other factors
 97 cannot determine it. The biological threshold test requires independent es-
 98 timates of the interaction kernel, its aggregation, and phase diffusion in the
 99 same cortical regime.

100 **3 Testing coherence through recovery after** 101 **awakening**

102 The proposed control variable is extracellular geometry. Sleep-associated
 103 changes in extracellular space and clearance (Xie et al., 2013) motivate a
 104 coupling parameter that evolves more slowly than phase dynamics. The
 105 extracellular volume fraction is approximately $\alpha = 0.2$ at diffusion tortuosity
 106 $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the cited sleep study reports a rise
 107 in α from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz cortical
 108 tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017).
 109 In an effective-medium description at fixed transmembrane current density,
 110 a larger extracellular volume fraction raises conductivity and reduces field
 111 amplitude.

112 This gives the coupling-gated hypothesis a directional prediction: contrac-
 113 tion of extracellular space on awakening raises E and, with the other factors
 114 in Eq. (4) held fixed, raises effective coupling. The proposed geometry–

115 coupling relation and its role in human recovery both require measurement.
 116 The effective-medium argument supplies a sign under stated conditions; cal-
 117 ibration supplies the magnitude and the mapping to the scalar model.

118 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 119 stationary branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

120 The crossing time satisfies $K(t_0) = K_c$. It is a prediction of latency only when
 121 coupling is measured independently of coherence. The subsequent shape
 122 and saturation depend on the later coupling trajectory. Critical slowing
 123 down restricts branch tracking near threshold, while finite populations have
 124 fluctuating order below it. A measurable onset therefore needs an explicit
 125 criterion above that fluctuation floor.

126 **3.1 What a recovery distribution can identify**

127 Equation (2) gives a concentration–coherence relation with no fitted shape
 128 parameter. Testing it requires separate estimators: r from circular resultant
 129 length and a from density shape. Estimating a by inverting R from the
 130 measured r makes agreement automatic. Even a valid agreement tests a
 131 statistical description shared by the von Mises family; it does not determine
 132 the coupling medium.

133 There is a stronger limitation. Successive equilibrated phase distributions

134 cannot, by themselves, distinguish three different recovery mechanisms. Let
 135 $q(t)$ be a prescribed ratio crossing 2, let $r(q)$ denote its stationary branch,
 136 and choose a fixed rate $D_0 > 0$. Increasing coupling with $K(t) = D_0 q(t)$
 137 and $D = D_0$ gives the same stationary density as fixed coupling K_0 with de-
 138 creasing diffusion $D(t) = K_0/q(t)$. Both have concentration $a = q(t)r(q(t))$.
 139 Uncoupled oscillators under a common prescribed drift $-h(t) \sin(\theta - \psi)$, with
 140 $h(t) = D_0 q(t)r(q(t))$ and $D = D_0$, have that same density. Their distribu-
 141 tions coincide window by window, even though their physical causes differ.
 142 Supplemental Material supplies the construction and its numerical feasibility
 143 check.

144 The missing observable is the physical clock. Resolved phase increments
 145 contain rate information: in the ideal continuous-path model, quadratic vari-
 146 ation accumulates at rate $2D$, while conditional increment means constrain
 147 drift. This can separate coupling and diffusion under the specified observa-
 148 tion model. Distinguishing endogenous coupling from the matched common
 149 drive additionally requires a measured drive or a controlled perturbation, be-
 150 cause their mean-field drifts can coincide. Identifying the coupling as electro-
 151 magnetic then requires evidence specific to that interaction. This hierarchy
 152 determines the experiment: distribution, dynamics and physical mechanism
 153 are distinct targets.

154 **3.2 Finite ramps change the observable onset**

155 Finite simulations characterize how the stationary prediction can appear in
156 a recovery measurement. The ramp study uses $N = 2000$, $D = 1$ and 32
157 replicas at four coupling speeds. Escape is the first of three consecutive
158 saved samples at $r \geq 0.2$ after threshold. The supplement gives parameters,
159 estimators and further controls.

160 Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape within the
161 fastest window. The three slower ramps have mean coupling delays 0.1947,
162 0.0684 and 0.0253. Their fitted onset exponents run from 0.898 to 0.417,
163 compared with 0.443 to 0.448 from the same estimator on the exact stationary
164 branch at the same coupling grids. At the fastest speed the ensemble mean
165 never reaches the declared fit window; the returned value 0.964 is a local
166 log-slope of a delayed trajectory.

167 This is delayed passage through a noisy bifurcation (Berglund & Gentz,
168 2002), with a direct consequence for the awakening test: the apparent expo-
169 nent depends on the ramp and on the fitting window. A deviation from the
170 square-root law constrains the adiabatic approximation only where branch
171 tracking is independently justified. The sweep does not bracket the specified
172 deviation-based critical ramp speed.

173 Additional controls test the approximations used by the physical model.
174 Time-step refinement attributes most of the sampled discrepancy from the
175 stationary order to integration bias. Population-size diagnostics show why
176 a shared random collective orientation can preserve unconditional pair cor-

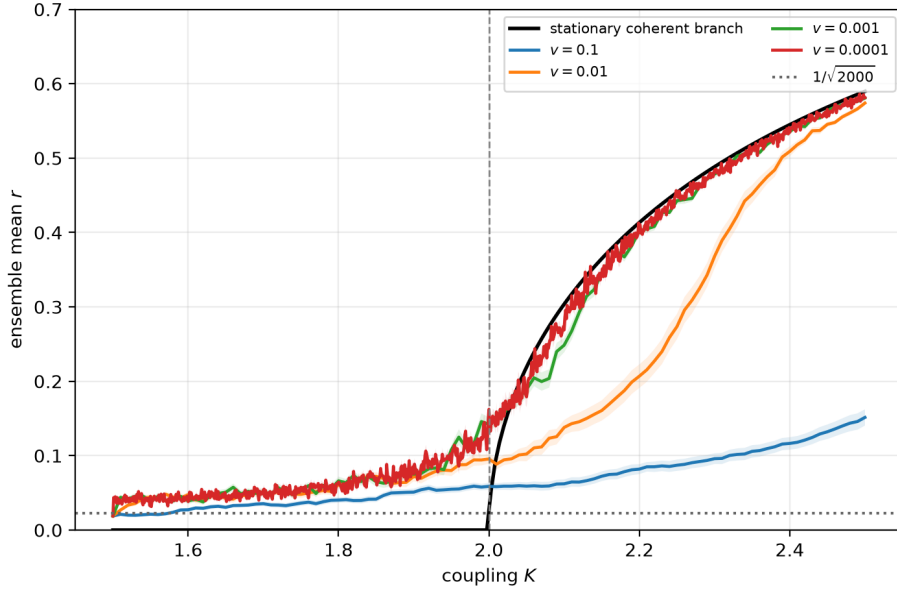


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

relation in the coherent regime. Spatial and signed-network controls exhibit coherence at specified total coupling, with the refined signed-network crossing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ still containing 2. These controls delimit finite regimes and numerical resolution; the supplement reports their full evidence. Calibrating the cortical coupling remains the experimental task.

3.3 What the estimators produce on EEG

Public propofol-sedation EEG (Bajwa et al., 2025) provides an exploratory exercise of the concentration and coherence estimators. The implemented

185 analysis pools channels over 100-ms windows, estimating a spatiotemporal
 186 phase distribution. Bipolar and common-reference montages produce differ-
 187 ent residuals. The bipolar results are compatible with the Bessel relation,
 188 but the concentration range is too narrow to distinguish it from its linear
 189 approximation at the observed scatter.

190 An informative recording needs a broader concentration range, estima-
 191 tor calibration that accounts for dependence, and an observation model for
 192 source mixing, referencing and temporal pooling. The sleep-recovery hypoth-
 193 esis remains untested by these sedation recordings.

194 **3.4 A protocol for the awakening experiment**

195 The analysis defines six requirements for the physical test.

- 196 **1. Coupling and diffusion in the same units.** Estimate K and D
 197 from resolved phase dynamics in the same cortical regime, separating
 198 diffusion, frequency heterogeneity and observation noise. A measured
 199 drive or controlled perturbation is additionally needed to distinguish
 200 endogenous coupling from the matched common-drive alternative.
- 201 **2. Geometry calibrated to coupling.** Establish the relation between
 202 an extracellular-geometry observable and effective coupling, with un-
 203 certainty. A diffusion-MRI or astrocytic proxy for volume fraction re-
 204 quires its own validation; the conductivity argument supplies only the
 205 expected sign.

206 **3. Adequate phase measurements.** For the proposed spatial estima-
207 tor, use at least 100 sites per estimate, r from resultant length, and
208 a from unweighted regression of log density on the cosine of centered
209 phase. This numerical minimum needs null calibration at the actual
210 sample count, concentration range and dependence. Temporal pooling
211 changes the estimand.

212 **4. An independent recovery endpoint.** Measure behavioral recovery
213 separately from the phase trace.

214 **5. A predeclared onset criterion.** Fix a coherence threshold above the
215 finite-size floor and account for finite ramp speed before interpreting
216 an onset exponent.

217 **6. Matched model comparison.** Compare coupling-gated recovery
218 with diffusion change, common drive and alternative slow-recovery
219 models under the same observation and filtering assumptions. Station-
220 ary distributions alone cannot resolve the equivalence in Section 3.1.

221 Failure of the concentration–coherence relation over an informative range
222 would constrain the phase-density model. Failure of the calibrated geometry–
223 coupling association would constrain the proposed physical gate. A com-
224 peting dynamical explanation that better predicts independently measured
225 recovery would count against its role in awakening. These tests address the
226 first requirement of the framework. Even a favorable outcome leaves the
227 question of what the coordinated activity represents.

228 4 Compatibility: from coordinated activity 229 to a global description

230 4.1 Why common timing is insufficient

231 Consider overlapping patches U and V , each carrying a local description.
232 Compatibility requires that restricting either description to $U \cap V$ gives the
233 same object. Agreement concerns the shared domain; descriptions on distinct
234 parts can remain different. For the finite spatial measures used here, equal
235 overlap restrictions give a unique measure on the covered domain. This is
236 the gluing property formalized by a sheaf.

237 In the cup example, suppose a visual description locates the cup on a
238 table and a reaching description relates the hand to the cup. To compare
239 them, a model must express their shared quantity, such as cup location, in
240 a common coordinate system. Restriction would retain that quantity while
241 dropping details peculiar to each description. Agreement on it permits the
242 visual description to contribute surface appearance and the reaching descrip-
243 tion to contribute hand configuration without duplicating the whole situa-
244 tion. Disagreement after coordinate conversion would make the two estimates
245 incompatible at that level of description, however closely timed their activity.

246 The formal counterexample makes the additional requirement visible. Ev-
247 ery patch has the same oscillator phase, yet two local mass profiles assign
248 different densities to a shared site. They cannot be restrictions of a single

249 global measure. Phase order therefore cannot establish compatibility. Con-
 250 tinuity of a physical field does not repair disagreement in the descriptions
 251 associated with it.

252 Agreement to within a tolerance determines less than a glued state, and
 253 more than nothing. Suppose the local profiles differ by at most ε in the
 254 uniform distance at every shared site. Nonnegative weights that vanish off
 255 their own patch and sum to one at each site then select a global profile lying
 256 within ε of every patch on that patch, and any two such selections differ by
 257 at most ε , with no dependence on how many patches cover a site; at $\varepsilon = 0$
 258 the selection is the unique glued state. What the tolerance costs is existence:
 259 for every positive ε there are families at that tolerance admitting no global
 260 description at all, so selection returns a choice among states rather than an
 261 extension of the data. A readout that is L -Lipschitz in the state varies by at
 262 most $L\varepsilon$ across selections, which is the resolution condition an interpretation
 263 at scale δ would have to meet. The supplement states these theorems and a
 264 three-site witness that attains the bound. Selection supplies no process by
 265 which local descriptions come to agree, and no rule that picks the cover or
 266 the weights.

267 For a theory of experience, the attraction of gluing is this contribution
 268 of different local contents to one description. The cup example does not yet
 269 instantiate the theorem: receptive territories and content domains must be
 270 related to the spatial patches, and their descriptions must have the required
 271 extension property. In particular, a decoder assigning probability laws to per-

ceptual contents needs a further extension model, because matching overlap
 marginals need not determine a unique global joint distribution (Abramsky &
 Brandenburger, 2011). The spatial-measure construction provides a precise
 case of gluing; a content model must justify its own existence and uniqueness
 claims.

4.2 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might appear to supply that missing representa-
 tional structure. The squared-drift control shows why the objective mat-
 ters. Symmetric coupling conserves mean drift: pairwise sine terms cancel
 when summed over nodes. Minimizing the sum of squared drifts at that
 fixed mean therefore rewards equalizing drift across nodes. In the chosen
 frequency-cluster model, this favors coupling between unlike-frequency clus-
 ters, whereas the diagnostic template rewards coupling within like-frequency
 clusters. The objective and the scored structure pull in different directions.

For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

The formalization constructs the bounded drift operator A and its adjoint,
 obtains $\nabla\sigma = (2/D)A^*(AK + \omega)$, and proves descent for the specified gradient
 flow. This is a squared-drift functional; identifying thermodynamic entropy

291 production would require probability-current information.

292 The finite control co-evolves noisy phases and symmetric, nonnegative
293 couplings at fixed total resource. The frequency-cluster template is a di-
294 agnostic covariance of shared latent signals, not a training input. Across
295 three seeds, second-half order remains between 0.9167 and 0.9222, while the
296 within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to
297 0.3507–0.6135 (Fig. 4). The supplement reports random-update, relabelled-
298 gradient and frozen-kernel controls, including comparisons at overlapping
299 deformation and coherence ranges and against partitions blind to frequency.

300 The objective reduction is transient: averaging over complete update
301 intervals removes the apparent gain seen immediately after updates. A pre-
302 specified grid of 6 rate–cadence combinations contains no case meeting the
303 declared 5% sustained-reduction criterion while retaining coherence and los-
304 ing alignment. The finding is coherent adaptation away from the specified
305 structure, with no demonstrated sustained cost reduction.

306 Kernel alignment and overlap compatibility are different observables.
307 This control tests the narrower suggestion that coherent adaptation under
308 this objective acquires the designated environmental structure. It fails for
309 an intelligible reason: the objective rewards a different organization. A
310 thermodynamically specified learning model can include data in its objective
311 (Corberi et al., 2026); explaining agreement about the cup would likewise
312 require a learning rule connected to the relevant content.

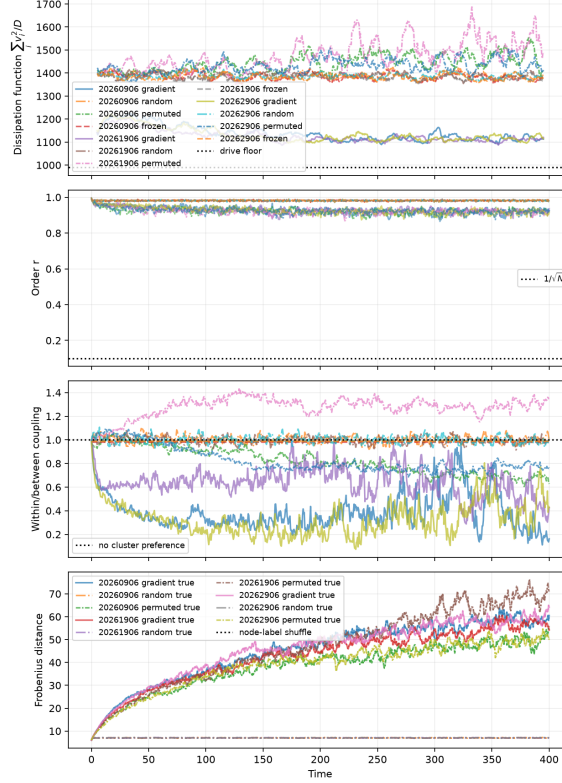


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

313 4.3 What must be specified and measured

314 Compatibility enters the construction as a hypothesis about the local states.
315 For deterministic positive coupling, convergence to a potential minimum is
316 proved under an explicit initial basin condition; the proposed bridge E78
317 connects supercritical noisy coherence to relaxation of the specified cover.
318 Neither step explains how initially inconsistent representations become con-
319 sistent. That is the mechanism a biological account must add.

320 A neural test must first identify the domains and objects whose agreement
321 is at issue. Two decoders should estimate a shared content on overlapping
322 domains, with restriction maps specified before inspecting their agreement.
323 Their discrepancy needs a null under decoding noise, and the procedure must
324 distinguish deliberately constructed compatible and incompatible examples.
325 A common prior can make separate decoders agree artificially; region-specific
326 bias can make them disagree artificially. The supplement develops these re-
327 quirements. An estimator built from phase agreement would miss the prop-
328 erty the counterexample separates.

329 Choosing a cover is also part of choosing a candidate subject. Trivial
330 descriptions can agree while carrying little content, and a suitably chosen
331 partition can conceal disagreement. The scientifically useful choice must
332 therefore be constrained independently by what local activity encodes and
333 which shared quantities can be compared. The uniqueness theorem then
334 answers a precise question about that choice: whether the supplied local
335 descriptions determine one global state. It leaves the empirical selection of

domains and descriptions to the model.

5 Self-representation: a state that models it-self

5.1 The reflexive construction

Compatibility yields a global state. Reflexivity adds a relation between that state and an encoding within it. Equip global sections with a complete metric. Under the additional restriction-resonance condition, the encoding of a section s is its restriction to a region U , and the map takes the form

$$F(s) = \text{readout}(s|_U). \quad (7)$$

The same region and readout are used across the model's state space. If F contracts and fixes the section s_* supplied by the cover, Banach's theorem makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 supplies the contraction law and $F(s_*) = s_*$ for the boundary's encoding-readout composite. Identifying that encoding with restriction is a separate resonance condition; E89 does not supply it. The supercritical order parameter provides an inequality used by the specified contraction bound; constructing the encoding and readout remains a separate obligation.

The equality ties the reconstruction to the state formed from the local

353 descriptions. In the cup example, the candidate global state would include
354 the represented scene and the perceiver’s relation to it. A local encoding
355 would have to retain enough information for the specified readout to recover
356 that modelled state. An encoding of cup position alone would be insufficient
357 if the readout could not recover the represented hand and viewpoint. The
358 term *predictive* names this reconstruction map; a temporal neural prediction
359 requires a dynamical interpretation.

360 **5.2 Why connect these properties to unity and a min-** 361 **imal self?**

362 We identify the section glued from compatible local states with the unity
363 of a conscious episode, and its accurate internal reconstruction with a mini-
364 mal self. Two features of the episode motivate this interpretation: different
365 contents belong to one situation, and that situation is experienced from a
366 perspective.

367 The unity proposal starts from a distinction between having several con-
368 tents and having several experiences. Seeing the cup’s surface and feeling
369 the reaching hand contribute different contents to one situation. Compat-
370 ibility expresses their agreement about what they share while preserving
371 those differences. An episode can also contain uncertainty or conflicting
372 judgments. A model might represent the cup’s apparent position separately
373 from a judged position and consistently describe the discrepancy. Deciding

374 whether two estimates concern the same quantity or distinct aspects of ex-
375 perience is therefore part of defining the content model. A choice made only
376 to ensure agreement would give the unity proposal no empirical force.

377 The motivation for a minimal self is that the cup is encountered from a
378 location and in relation to a body that can reach it. The proposed internal
379 representation concerns the perceiver’s condition in that encounter: where
380 the perceiver is situated, what is being represented and how the body relates
381 to it. Such a representation could be implicit in the organization of activity;
382 it need not describe a person verbally or carry an autobiographical narrative.
383 Requiring it to reconstruct the same global state links the perspective to the
384 episode whose contents it organizes.

385 The fixed-point condition requires accuracy about the very state contain-
386 ing the encoding. Reconstruction alone is insufficient for a perspective: a
387 recording can be reconstructed without being used by a system to represent
388 its own condition. The proposed interpretation therefore needs evidence that
389 the encoding helps the system distinguish changes in itself from changes in
390 its surroundings. Changing the encoded hand or viewpoint while holding
391 the external scene fixed should alter the system’s estimates of its relation
392 to the cup as the readout predicts. This functional criterion goes beyond
393 $F(s_*) = s_*$; whether the represented perspective is experienced remains a
394 further question.

395 Accuracy concerns the variables represented by s_* , rather than every mi-
396 croscopic physical fact. For a fixed map F with Lipschitz factor $q < 1$ and

397 fixed point s_* , approximate reconstruction has a quantitative bound:

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (8)$$

398 At the stipulated rate $q = e^{-(K-2D)\tau/2}$, this radius grows as $2\varepsilon/((K-2D)\tau)$
399 near threshold, for fixed positive ε and τ . The bound weakens while the
400 exact fixed point remains unique. The supplement gives explicit inequalities
401 and composes reconstruction error with displacement of the state. A cortical
402 realization must specify the represented variables, their metric and the
403 required fidelity; changing readouts require a further dynamical model.

404 We propose reflexivity as necessary because even an unreflective episode
405 has a perspective: the cup appears somewhere relative to the perceiver,
406 whether or not the perceiver attends to themselves. Requiring internal self-
407 encoding on that basis is the philosophical hypothesis. Gluing alone does
408 not require it; the dissociations below test the two commitments separately.

409 5.3 Which cover and which self-model?

410 Both uniqueness results are conditional on their objects. Gluing gives one
411 global extension of specified compatible local data; Banach gives one fixed
412 point of a specified contracting map. These solve ambiguity within a chosen
413 construction. They do not choose between different covers, different encod-
414 ings or different candidate subjects. Integrated information theory makes
415 exclusion an explicit part of its account (Tononi et al., 2016); the present

416 uniqueness results do not settle that selection problem.

417 The relevant cover and map must be tied to neural organization by evi-
418 dence independent of the desired conclusion. Local descriptions should pre-
419 dict the contents attributed to their domains, and their restrictions should
420 preserve meaning on shared domains. The encoding and readout should be
421 specified before testing them, with their sensitivity to changes in the pro-
422 posed state measured over a declared range.

423 A constant readout returning one predetermined state contracts and has a
424 fixed point, even when it discards an encoding supplied by restriction. Such a
425 construction fails the sensitivity requirement above. This is why selecting the
426 map requires evidence about its functional role in addition to mathematical
427 existence.

428 **5.4 Dissociations that would bear on the identification**

429 A test of the full commitment must measure its clauses separately: effective
430 coupling relative to diffusion, overlap compatibility, and the accuracy of the
431 proposed self-encoding, alongside an independent endpoint for experience.
432 A case meeting all three physical and representational conditions with evi-
433 dence against a unified conscious episode would challenge their sufficiency.
434 Unified experience with demonstrably incompatible local descriptions would
435 challenge the unity identification. Unified experience with compatibility but
436 without the specified self-encoding would challenge the reflexive requirement.

437 Anesthetic emergence, dreaming and dissociative states are candidate set-

438 tings because the framework asks whether coordination, unified content and
 439 self-representation can separate. Interpreting such evidence requires a de-
 440 clared relation between the experience endpoint and report; absence of report
 441 alone cannot define absence of an episode. The compatibility observable in
 442 the supplement is a specification, and the corresponding neural self-encoding
 443 has yet to be identified. These are the main obstacles to testing the inter-
 444 pretive commitment.

445 The proposal concerns which state is conscious. It supplies no explanation
 446 of why satisfying these relations should be accompanied by experience. Nor
 447 do its gluing and fixed-point conditions select an electromagnetic substrate:
 448 the cortical field commitment resides in the physical identification and cali-
 449 bration. Their multiple realizability leaves the status of other biological and
 450 artificial implementations open.

451 **6 Thermodynamic constraints and the con-** 452 **ditional composition**

453 The broader composition constrains one proposed physical implementation
 454 through finite-register capacity $H(p) \leq \log |X|$, erasure heat under a specified
 455 system–bath model (Landauer, 1961; Norton, 2005; Shenker, 2000), and a
 456 separate field-boundary result for disconnected vacuum components (Kibble,
 457 1976). The supplement develops these results and their witnesses. The phase
 458 threshold and representational argument stand independently of a derivation

459 from finite capacity.

460 The constraint most directly relevant to representation concerns predic-
 461 tive memory. For a state X_t driven by an environmental signal S_t ,

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (9)$$

462 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, the bracket is
 463 nonnegative (Still et al., 2012). The bound limits nonpredictive memory at a
 464 given dissipation budget; zero retained information satisfies it too. Relating
 465 it to Landauer heat requires that memory waste be bounded by erased en-
 466 tropy at the same thermal scale and heat budget. These distinct information
 467 constraints (Rosas et al., 2026) leave content and learning dynamics to be
 468 specified.

469 An acting agent changes its future inputs, so the passive assumption
 470 requires justification for the variables and interval considered. A feedback
 471 model specifies a policy $A_t \sim \pi(\cdot | X_t)$, world dynamics $S_{t+1} \sim K(\cdot | S_t, A_t)$,
 472 an observation channel and a memory update. For finite states, the exact
 473 information balance is

$$I(X_t; S_t) - I(X_t; S_{t+1}) = I(X_t; S_t | S_{t+1}) - I(X_t; S_{t+1} | S_t). \quad (10)$$

474 Both conditional terms are nonnegative. The last term vanishes for the
 475 passive model; with feedback, the signed difference can be negative because

476 actions create correlations with the future. Its causal interpretation requires
 477 an environmental state sufficient to account for the relevant dynamics.

478 The finite thermodynamic extension instead bounds joint entropy reduc-
 479 tion. For an elementary update that holds the controller fixed, strictly
 480 positive forward and reverse path probabilities give $\Sigma = D_{KL}(P_F||P_R) =$
 481 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with Q positive
 482 into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa, 2013). This
 483 supplies the thermodynamic constraint for the active branch below. A proved
 484 two-substep extension composes actuation with a memory update initial-
 485 ized from its coordinate-swapped final law. It identifies the output with the
 486 perception–action channels and cancels the same intermediate entropy and
 487 expected energy. At a common thermal scale, the whole update’s joint en-
 488 tropy decrease is bounded by its total heat budget divided by $k_B T$, without
 489 assuming stationarity. The supplement gives the balances and reciprocal
 490 two-bit witnesses with explicit interaction energy, heat and work. Action
 491 and observation alias the existing states; additional physical registers require
 492 further accounting.

493 An explicit objective also permits finite policy selection. For a supplied
 494 terminal reward and a finite set of deterministic policies, a nonempty set sat-
 495 isfying the complete update’s heat budget has a reward maximizer. A two-bit
 496 lamp task has a unique feasible optimum that strictly improves on a feasi-
 497 ble constant-action baseline; a stronger policy exceeds the budget. Reward,
 498 heat and work are evaluated on each policy’s actual composed process, with

499 common initial law, channels, energy and reservoir convention. The base-
500 line meets the heat budget while missing the task target, so the objective
501 supplies content that thermodynamics alone does not. This result compares
502 fixed policies for one update.

503 A separate finite learning model specifies how a policy register adapts. Its
504 bit selects between the two feasible lamp policies, and its energy decreases
505 with their known expected reward. An autonomous thermal channel changes
506 the register, carrying each final distribution into the next update. From the
507 specified initial preparation, Lean proves strictly increasing expected task
508 reward and geometric convergence to a stationary mixture. Transition heat
509 obeys local detailed balance, and cumulative update heat equals the regis-
510 ter’s finite energy loss at zero external work. The mixture remains below
511 the optimal policy; individual updates can lower reward, and a different
512 initial preparation lowers expected reward. This is adaptation toward an
513 encoded objective. Learning unknown rewards from observations, preparing
514 the register and implementing its readout in an actuator require additional
515 accounting. Task reward and task heat are prospective expectations under
516 the prepared lamp law; the learning clock executes no task episodes and
517 supplies no spatial coupling convergence or cortical identification.

518 6.1 What the composition establishes

519 Table 1 lists the connecting hypotheses of two branches. The theorem **chain**
520 uses passive predictive memory; **chain_active** uses a specified finite feedback

521 process. Each accepts eight implications and shares the passage from coarse-
 522 graining to the phase model and its proposed representational role. In both,
 523 the finite register and the field boundary are independent starting objects.

524 For the active branch, the controller has the register's state type and the
 525 fourth node records

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (11)$$

526 where p, p' and $\langle Q \rangle$ belong to that same controlled substep. E34Active sup-
 527 plies positive path probabilities, local detailed balance at the register's ther-
 528 mal scale θ , and the allocation $\langle Q \rangle \leq B$, with B equal to the register's
 529 heat. Landauer's lower bound does not imply this upper-budget allocation
 530 or identify erasure with actuation. A resource model of one register supplies
 531 it instead: every operation is controlled by that register and exchanges heat
 532 with its reservoir at θ , and the register's dissipated heat is the total its op-
 533 erations deliver there. An operation that does not increase the joint entropy
 534 then has a nonnegative share, by the left inequality of Eq. (11) applied to it,
 535 so an operation whose companions all compress has a share no larger than
 536 the register's dissipation. That accounting identity and that compressiveness
 537 are the model's physical inputs; the allocation follows from them. Composing
 538 the operations into one evolving law with its preparation and power source,
 539 and deriving the identity from a microscopic bipartite dynamics, are separate
 540 models. E45Active requires convergence of the specified coupling energies.

541 One scalar constitutive model connects those energies to the named feedback
 542 step: writing $d = H(p) - H(p')$, it sets the spatial density to $f_A(x) = g d b(x)$,
 543 with supplied gain $g \geq 0$ and profile b . Regular spatial refinement gives

$$E_n(f_A) \longrightarrow L_A = g d \int_S b d\mu, \quad L_A \leq g \frac{B}{\theta} \int_S b d\mu \quad (12)$$

544 when $\int_S b d\mu \geq 0$. The limit follows from mesh assumptions; the upper
 545 bound uses Eq. (11) for the same process. The constitutive law uses ensemble
 546 entropy reduction and supplies neither a microscopic actuator nor its work
 547 account. Changing the gain or profile at fixed process and heat changes the
 548 coupling energies.

549 A second, microscopic model supplies an actuator and a kernel. The
 550 hardware carries finitely many spatial response modes with fixed profiles ϕ_i
 551 and installation prices u_i . A configuration z of the controller and its switches
 552 sets the mode occupancies $c_i(z)$, and those same occupancies determine both
 553 a reciprocal kernel on the product space and a stored installation energy,

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (13)$$

554 A transition $z \rightarrow z'$ changes K only through the modes whose occupancy it
 555 changes, so a pair outside the response of those modes is untouched, and the
 556 transition's external work is $U(z') - U(z) + Q$. The kernel the executed step
 557 installs is the average of K_z over the law that step actually leaves; its change is
 558 the mean of the pathwise changes, and the mean of the transition works splits

559 into the change in stored energy and the channel’s heat. A finite measurable
560 partition of the substrate with one sample per cell reconstructs the installed
561 kernel throughout each cell pair, and the cell-mass weighted finite energy
562 equals the product-measure integral of that reconstruction exactly. On a
563 compact substrate with continuous profiles, a sample error tending to zero
564 gives

$$\mathcal{E}_n \longrightarrow \iint K d\mu d\mu. \quad (14)$$

565 The refinement index here is spatial and is independent of the number of ex-
566 ecuted control updates; convergence in one is not convergence in the other.
567 The same hardware discharges the passive edge from a declared predictive
568 system’s own joint law, and two declared systems with different dissipated
569 work give the identical edge, so predictive efficiency selects neither the hard-
570 ware nor the partition. Mode profiles, prices, occupancy readout, reservoir
571 convention and substrate are declared hardware data, which no heat bound
572 supplies. Fabricating that hardware, powering it beyond one prepared up-
573 date, and identifying the substrate or the modes with a physical field remain
574 separate requirements.

575 Two structures also enter both theorems: a thermodynamic cover, in-
576 cluding overlap compatibility, and a reflexive boundary specifying the local
577 encoding and readout. Given these and the connecting hypotheses, both
578 conclude **UnifiedSelf**: the section glued on that cover is the unique fixed
579 point of self-prediction. Restriction resonance, needed for the interpretation

Connection	Status	Required assumption
E12	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity and the boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: a predictive structure with the register’s thermal scale and heat budget, and nonzero wasted memory. Active: positivity, local detailed balance and the register’s heat allocation for the named elementary process, the allocation being derived from a ledger of that register’s own operations. The ledger identity and the compressiveness of the register’s other operations are that model’s inputs.
E45 / E45Active	Modelling assumption	Passive: a convergent regime, for hardware holding a declared predictive law. Active: Eq. (12) from a supplied gain, profile and meshes; Eq. (14) from declared response modes.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence drives a supplied cover to its relaxed equilibrium at coupling at least the named mean-field value.
E89	Modelling assumption	A map mediated by local self-encoding fixes the section glued on that cover and has the Lipschitz factor $q < 1$ used in the exact and approximate-state results.

Table 1: Connecting hypotheses of the passive and active compositions, eight per branch. They share E12, E23 and E56–E89. The cover’s compatibility is supplied as a structure field. Identifying the boundary’s encoding with restriction requires a separate resonance condition. The mathematical objects and node-level results are mapped in Supplemental Material, Table S1.

580 in Eq. (7), is a separate condition. The result binds the fixed point to the
581 specified local states; compatibility and accuracy at that state are premises,
582 while uniqueness follows from the construction.

583 The joint witnesses combine a double well, a register, a changing uniform-
584 grid mesh and a three-site cortex; the active branch includes the noisy two-bit
585 actuator. They establish that the hypotheses can hold together in distinct toy
586 components. Mesh convergence is supplied independently of the thermody-
587 namic bounds, and no common biological mechanism is established. A second
588 active witness composes the same branch with the bridge discharged from
589 that register’s ledger: its two operations deliver $(2/5) \log 2$ and $(3/5) \log 2$ to
590 one reservoir, summing to the register’s $\log 2$. A third active witness closes
591 one spatial response mode with a thermal switch: executing the channel
592 strictly raises a pairwise kernel value, closing the mode costs its stored en-
593 ergy plus the channel’s heat, and uniform partitions of an atomless interval
594 give a coupling energy that changes between the first two refinements and
595 converges to the value the field witness uses. The active witnesses reject
596 a zero heat budget, an incompatible energy limit, a compressive operation
597 above the register’s whole dissipation, a ledger whose other operation draws
598 heat out of the reservoir, an uncharged installation, and a wrong spatial limit
599 for the microscopic sequence. Counterexamples also expose the role of over-
600 lap agreement. The development declares no additional Lean axioms beyond
601 the foundational axioms used by its libraries; physical premises are explicit
602 hypotheses and model fields.

603 7 Discussion

604 The strongest results concern what a coherence measurement leaves unre-
605 solved. Different recovery mechanisms share stationary distributions; equal
606 phases permit incompatible local descriptions; and the tested plasticity ob-
607 jective sustains coherence while moving away from its diagnostic structure.
608 These findings make the coupling hypothesis more demanding to test and
609 prevent a successful synchronization model from doing unexplained repre-
610 sentational work.

611 The proposed account identifies a compatible, internally represented state
612 with a conscious episode. Its main explanatory gap is the mechanism that
613 would maintain those relations: compatibility and accuracy at the glued state
614 enter the construction as premises. Their mathematical consequences leave
615 open why cortical activity should acquire or preserve them.

616 The most informative next step for that connection is a test of whether
617 coordination helps maintain independently measured content agreement. Af-
618 ter validating separate decoders and their overlap restrictions, a controlled
619 change in coupling or coordination could test whether agreement deterio-
620 rates in the predicted way, with controls for changes in sensory information
621 and decoder quality. Such a result would bear directly on the proposed role
622 of coherence. In parallel, the awakening hypothesis needs its own calibrated
623 geometry–coupling relation and resolved phase dynamics to distinguish a cou-
624 pling increase from diffusion change or common drive. A favorable recovery

625 trace alone would leave the representational connection open.

626 Testing the self clause requires perturbing a candidate encoding and com-
627 paring the resulting changes in represented scene, body and viewpoint with
628 the readout’s predictions. Establishing that functional role would make dis-
629 sociations from unified experience informative. The exact fixed-state ideal-
630 ization must also be assessed against the fidelity and time dependence of the
631 biological process.

632 Integrated-information, global-workspace and predictive-processing ac-
633 counts organize unity, access and modelling differently (Tononi et al., 2016;
634 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal
635 contributes a candidate interaction mechanism and explicit relations to test
636 between local content and internal self-encoding. Its explanatory value will
637 turn on whether those relations can be identified independently and shown
638 to matter for experience. For the cup example, that means explaining how
639 the system maintains one represented situation and how the perceiver’s place
640 within it contributes to the episode.

641 **Data and code availability**

642 The Lean development, simulation scripts, tests and saved numerical sum-
643 maries are maintained in the manuscript source repository. Publication sim-
644 ulation values are regenerated from saved summaries without rerunning pro-
645 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset

646 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
647 in Supplemental Material. The source repository contains the reproducible
648 analysis and does not redistribute the raw recordings.

649 **Funding and competing interests**

650 This research received no external funding. The author declares no compet-
651 ing interests.

652 **Use of AI tools**

653 AI assistance to this work was substantial and is not separable by task.
654 Several coding assistants and chat interfaces were used in an interleaved way
655 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
656 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
657 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
658 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
659 they contributed to the Lean development, the simulation and analysis code,
660 and the drafting and revision of both documents. Because the tools were
661 used in combination and in alternation, no section, proof, figure or result is
662 attributable to any one of them, and none is claimed to be. The author is
663 responsible for all content, including everything a tool produced. Machine
664 checking establishes the stated Lean results under their hypotheses; it does

not validate biological interpretations or replace scientific review of generated
code and prose.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
electric fields emerge from and allow representational drift. *NeuroImage*,
253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
tion theory: from consciousness to its physical substrate. *Nature Reviews*
Neuroscience, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
tational theory of conscious processing. *Current Opinion in Neurobiology*,
25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
Reviews Neuroscience, 11(2), 127-138.
- Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.

683 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
684 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
685 *Progress in Neurobiology*, 226, 102465.

686 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
687 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
688 9(8), 45-60.

689 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
690 testable hypothesis about the characteristics of conscious as opposed to
691 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

692 McFadden, J. (2020). Integrating information in the brain's EM field: the
693 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
694 niaa016.

695 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
696 guide neocortical network activity. *Neuron*, 67(1), 129-143.

697 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
698 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
699 95-103.

700 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
701 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.

702 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
703 on brain circuits in rats and humans. *Nature Communications*, 9, 483.

704 Verkhatsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
705 *logical Reviews*, 98(1), 239-389.

706 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
707 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
708 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

709 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
710 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
711 ity of complexity measures to capture dreaming during propofol sedation.
712 *Scientific Reports*, 15, 32012.

713 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
714 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
715 brain. *Science*, 342(6156), 373-377.

716 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
717 *Physiological Reviews*, 88(4), 1277-1340.

718 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
719 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

720 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
721 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
722 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

723 Landauer, R. (1961). Irreversibility and heat generation in the computing
724 process. *IBM Journal of Research and Development*, 5(3), 183-191.

725 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
726 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

727 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
728 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

729 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
730 of prediction. *Physical Review Letters*, 109(12), 120604.

731 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
732 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

733 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
734 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
735 031015.

736 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
737 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
738 PhysRevLett.111.180603.

739 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
740 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.

741 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
742 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
743 181.

- 744 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
745 archive.pitt.edu/115.
- 746 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
747 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
748 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 749 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
750 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
751 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 752 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
753 bifurcations with additive noise. *Probability Theory and Related Fields*,
754 122, 341–388. doi:10.1007/s004400100174.
- 755 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
756 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
757 doi:10.1088/1367-2630/13/11/113036.
- 758 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
759 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
760 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
761 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- 762 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
763 terns underlie complex brain dynamics and are related to cognitive pro-

764 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
765 023-01626-5.